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IN FOCUS

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U.S. Satellite Capabilities for Tracking the Wildfire Life Cycle

Satellite capabilities support activities at virtually every stage of the wildfire life cycle: characterizing the prefire environment (e.g., seasonal climate predictions, vegetation); detecting and monitoring wildfires and smoke; and assessing postfire hazards (e.g., debris flows, flash floods) and ecosystem recovery. The National Oceanic and Atmospheric Administration (NOAA) notes that satellites "collect valuable information about a fire's location, size, temperature and intensity, which is difficult or impossible to gather" through ground-based sensors, aerial drones, or aircraft. These approaches may provide complementary data to satellites, particularly when satellite data are limited in spatial or *temporal* (i.e., the time between observations of the same area) resolution, and given the money and time needed to develop and deploy new satellites.

NOAA, the National Aeronautics and Space Administration (NASA), and the U.S. Geological Survey (USGS) collect satellite observations across the wildfire life cycle, including detecting and monitoring wildfires in *near real-time* (NRT; i.e., data available within about one to three hours). Congress has considered legislative proposals aimed at improving satellite observations for wildfire purposes (e.g., H.R. 527, H.R. 753, H.R. 5782/S. 91 in the 119th Congress). This CRS In Focus describes current and planned U.S. civil satellite observation systems that may contribute data relevant to the wildfire life cycle.

Satellite Wildfire Applications

Satellites can collect environmental information about fire weather and vegetation conditions that contribute to increased wildfire risk. These pre-wildfire observations, among others, guide predictions of the likelihood and intensity of wildfire outbreaks. Land managers and scientists can use post-wildfire satellite observations to assess the severity, extent, and impact of a wildfire. The assessments may support postfire hazard risk assessments and inform efforts to monitor landscape recovery.

Satellites can be used to detect wildfires, generally in concert with nonsatellite wildfire detection techniques. Some satellites can make NRT observations to support early wildfire detection, which increases the likelihood of timely response. First responders depend on NRT observations to plan firefighting operations and to inform firefighters' situational awareness during the wildfire. To be useful for these purposes, satellites must provide timely data with the necessary frequency and spatial resolution to users on the ground. Higher spatial resolutions would allow for the detection of smaller, lower-temperature wildfires. *Data latency*, the total time between data acquisition and public availability, may vary across satellite instruments (even within the same satellite) and may affect the use of observations for wildfire applications.

Both geostationary orbit (GEO) and low-Earth orbit (LEO) satellites provide wildfire-related observations but do so at different spatial and temporal resolutions. GEO satellites, which orbit at higher altitudes, remain over the same location on the Earth and provide continuous coverage of a large area but at a lower resolution relative to LEO satellites. LEO satellites, which orbit at lower altitudes, provide a more limited field of view while allowing for relatively higher spatial resolution. LEO satellites complete several orbits per day; each orbit allows data collection for different locations and times of day, but observations at each location have a lower temporal resolution (less frequent observations). To achieve similar spatial or temporal coverage as a GEO satellite, multiple LEO satellites (i.e., a *constellation)* may be needed. Coupling LEO and GEO data may provide a more comprehensive approach to tracking the wildfire life cycle.

NOAA GEO Satellites

NOAA's Geostationary Operational Environmental Satellites (GOES)-R Series, a two-satellite operational system, hosts the Advanced Baseline Imager (ABI), which produces a scan of the Earth every 60 seconds with a spatial resolution of 500 to 2,000 meters. ABI's data help experts detect smoke plumes and heat signatures to identify and respond to fires in NRT. Monitoring smoke in NRT is useful in directing firefighting efforts and protecting public health. GOES-R Series satellites also host a lightning mapper (GLM), which collects data on the frequency, location, and extent of lightning, a common wildfire ignition source.

NOAA anticipates its future GEO satellite mission, Geostationary Extended Observations (GeoXO), will include a new imager (GXI) that is more advanced compared with ABI. GXI may detect a wildfire a quarter of the size of those ABI can detect while providing data to first responders more quickly. NOAA aims to launch the first GeoXO satellite carrying GXI in 2034. In 2026, the Trump Administration "discontinued" the procurement of a GeoXO lightning mapper (LMX) for budgetary reasons. LMX would have been an improved version of GLM. For more information, see CRS Report R49026, *NOAA's Geostationary Extended Observations (GeoXO) Satellite Program and Issues for Congress*.

Selected U.S. LEO Satellites

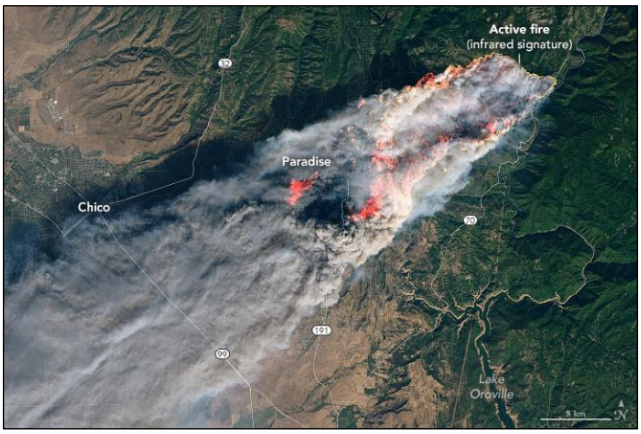
NOAA's Joint Polar Satellite System (JPSS), a fleet of three LEO satellites, provides global observations twice a day. JPSS satellites host the Visible Infrared Imaging Radiometer Suite (VIIRS), which has a 375 meter spatial resolution. VIIRS also has a Day-Night Band that enhances the detection of nighttime wildfires. VIIRS wildfire observations have a data latency of about 50 seconds—sometimes referred to *ultra real-time*—a speed that rivals

nonsatellite wildfire detection methods. NOAA plans to launch one of its future polar-orbiting satellites, JPSS-4, in 2027 with the current generation of VIIRS.

NASA's Terra satellite hosts instruments used for wildfire detection and response. NASA expects Terra to cease operations by April 2027 and has encouraged users to use data products from JPSS's VIIRS. Some Members of Congress have raised concerns over the pending transition.

Under the Sustainable Land Imaging Program, NASA develops and launches Landsat satellites and USGS operates the satellites and manages their data. Landsats 8 and 9 host the Operational Land Imager (OLI) and the Thermal Infrared Sensor (TIRS)—both instruments have been used for wildfire-related applications (**Figure 1**). Information collected by Landsat helps users understand wildfire risk and ecosystem recovery, detect wildfires, and map burned areas following a wildfire to assess risk of post-wildfire debris flows and monitor landscape recovery. While Landsat observations generally have a data latency of four hours or more, a Landsat Fire and Thermal Anomalies product provides OLI data for wildfire detection within 30 to 60 minutes. NASA and USGS are developing a follow-on "super-spectral" Landsat mission to launch in 2031 that is anticipated to support improved wildfire applications. The Biden Administration initially proposed a three-satellite constellation (Landsat Next), which the Trump Administration restructured to Landsat 10, comprising a single satellite. A single-satellite mission would reduce costs. It would lower temporal resolution compared with a three-satellite mission, while spatial resolution would remain the same.

Figure 1. Landsat 8 Image of the Camp Fire near Sacramento, CA, November 8, 2018



Source: NASA Earth Observatory.

The U.S.-India NASA-ISRO Synthetic Aperture Radar (NISAR) mission provides information about vegetation, fuel condition, and soil moisture over the entire planet every 12 days with a 200 meter resolution. NISAR also provides observations at times when imagers are ineffective (e.g., during cloudy conditions or at nighttime).

Issues for Congress

New satellite missions pose challenges, including the need for multiyear appropriations and the inherent risks of developing new technology. Conversely, satellite retirements may pose data continuity challenges. At issue for Congress is weighing the potential cost savings of retirement or otherwise reduced federal observations against maintaining or advancing data temporal and spatial resolution for federal agency wildfire objectives.

Congress may continue its oversight of whether and how future federal earth observation satellites may contribute to tracking and fighting wildfires. For example, in March 2026, NASA announced its new Explorer for Artemis Geology, Lunar, and Earth (EAGLE) program, a series of satellites used to study the Earth and Moon. NASA expects to launch the program's first satellite in 2028 and its second satellite two to three years later. Both satellites could be used to detect and monitor wildfires.

To reduce federal spending, the Trump Administration has proposed to modify some future satellite missions by including current rather than improved instruments in certain missions or eliminating instruments or satellites from missions (e.g., GeoXO and Landsat). The proposed modifications may impact wildfire forecasting and response. Congress could direct some potential cost savings toward other wildfire-related activities. Congress also may consider potential consequences associated with the elimination of instruments and reduced satellite coverage.

Congress also may consider alternative ways to obtain satellite observations for wildfire purposes. Federal partnerships with U.S. commercial providers may help fill wildfire data gaps. In July 2026, NOAA entered into a $2.51 million contract with a California-based company to buy data for a Wildfire Imagery Pilot via the agency's Commercial Data Program. Some Members have proposed legislation related to NOAA purchasing wildfire data from the commercial sector (e.g., S. 306 and S. 3923 in the 119th Congress). Alternatively, Congress could direct federal agencies to review the potential applications of existing, in-orbit federal instruments not initially designed for wildfire use. For example, scientists proved an "unexpected" capability to monitor global changes in vegetation using an instrument on NASA's Plankton, Aerosol, Cloud, and ocean Ecosystem (PACE) satellite—which focuses on monitoring ocean health, air quality, and climate change. Other PACE instruments have been used to study wildfire smoke particulates.

Policymakers may continue to consider interagency efforts to disseminate wildfire data to federal and nonfederal response agencies. For example, the Fire Information for Resource Management System, a joint effort between NASA and the U.S. Forest Service, uses data collected from various satellites to provide NRT detection of wildfires in the United States and Canada. Alternatively, Congress has considered creating a pilot program to test and deploy technologies for wildfire prevention, detection, communication, and mitigation (e.g., S. 140 and S. 1462 in the 119th Congress).

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